XR-Robot Receptionist Agents: A User Study on Social Acceptance

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***Abstract*—Receptionist Agents (RAs) deployed in public environments must ensure effective user-agent interaction, safe navigation, and compliance with social norms. While standalone embodied robots struggle to balance interaction and mobility, agents based on eXtended Reality (XR) offer richer interaction but lack physical embodiment, limiting their ability to provide spatial guidance. To address these challenges, this paper proposes an XR-Robot RA architecture that distributes the receptionist role across two coordinated components: an interaction-centric XR agent responsible for user identification and engagement via dialogue, and a mobility-centric robot responsible for user guidance via physical navigation, enabling a clear separation between communication and navigation tasks.**

The proposed system is validated through a user study conducted in a real-world technological center. Social acceptance and user perception are evaluated using a structured questionnaire built upon an extended Almere model, while two robot navigation strategies are compared to assess the impact of motion behavior on perceived interaction quality. The results indicate generally positive user perceptions across all evaluated attributes, with statistically significant improvements in perceived adaptiveness and social presence for the advanced navigation strategy, highlighting the effectiveness of hybrid embodiment and separation of functional roles to enhance social acceptance of RAs.

***Index Terms*—Receptionist agent, service robotics, eXtended Reality (XR), social acceptance.**

I. Introduction

In recent years, the deployment of service robots [1] in public and human-shared environments has received increasing attention, supported by advances in robotics, Artificial Intelligence (AI), and Human-Robot Interaction (HRI). Within the domain of service robotics, Receptionist Agents (RAs) represent one of the most socially demanding application areas, as their operation requires simultaneous handling of autonomous navigation, verbal and non-verbal communication, and socially and contextually appropriate interaction with users. Typical deployment contexts include museums [2], airports [3], hospitals [4], and convention centers [5]. In these settings, interaction occurs with heterogeneous user populations, varying in age, gender, cultural and linguistic background, prior experience with robotic systems, and levels of physical or cognitive ability, within semi-structured environments, i.e., spaces characterized by a predefined spatial layout combined with partially predictable human presence and dynamic social activity [3]. The core RA functions include welcoming visitors, providing contextual information, performing user identification or authentication, and accompanying users to designated destinations [6]. Effective execution of reception tasks requires an integration of perception [7], dialogue management [8], and autonomous navigation capabilities, together with compliance with social norms, functional requirements, and ethical constraints [9].

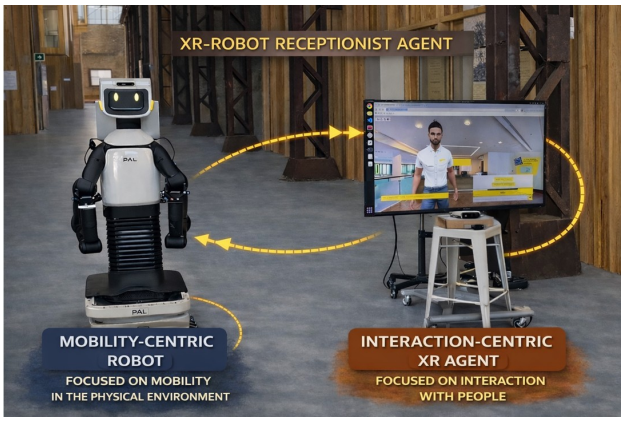

Fig. 1. Experimental setup of the proposed XR-Robot RA architecture operating in a technological center.

Existing RA implementations predominantly rely on single embodied robotic platforms responsible for interaction and physical navigation [10]. Although unified robotic embodiments enable direct engagement within shared spaces, several limitations arise in crowded or operationally complex environments, such as airports during peak hours, hospital corridors with continuous staff movement, or exhibition halls hosting simultaneous events. Interaction processes may exhibit an interplay with navigation requirements, leading to compromises between communication quality and motion safety. In addition, the expressive potential of physical robots is often limited by hardware constraints, sensing accuracy, or safety regulations [3]. Considerations of this nature motivate the investigation of alternative RA architectures in which interaction and mobility roles are distributed across complementary components while maintaining coherence across system functionalities.

Alongside developments in service robotics, eXtended Reality (XR) technologies, which encompass the reality-virtuality continuum, including Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR) [11], have undergone rapid

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technological progress and increased accessibility. Within this technological landscape, XR-based agents and conversational avatars have demonstrated strong potential for enhancing human-machine interaction by enabling immersive, multimodal, and context-aware experiences. XR agents can support natural language dialogue, expressive non-verbal communication, and adaptive interaction behaviors [12], [13], features that are particularly relevant for socially-rich reception scenarios requiring sustained engagement with users.

Building on the complementary strengths of XR agents and mobile robots, the present work proposes an XR-Robot RA architecture in which the receptionist role is distributed between two cooperating embodiments with distinct responsibilities. An interaction-centric XR agent handles user engagement, dialogue, and identification, while a mobility-centric robot performs navigation and physical guidance. Coordination enables coherent behavior by decoupling interaction from navigation constraints, allowing safe and socially compliant motion alongside flexible interaction. The hybrid configuration supports robust and scalable reception services across digital and physical domains in real-world public environments.

In addition to technical functionalities, the evaluation of social acceptance represents a fundamental requirement for XR-Robot RAs, given operation in open, public spaces and direct interaction with heterogeneous user populations. Foundational definitions describe social acceptance as a technology's ability to be introduced into various social contexts without causing discomfort or alienation among users [14]. Despite the relevance of earlier perspectives, they fail to capture the interaction complexity associated with XR-based agents operating in socially shared spaces. In the context of XR-Robot RAs, social acceptance can be more appropriately defined as the system's capability to support interactions that are satisfactory, transparent, safe, secure, and trustworthy, while remaining aligned with the social norms and situational expectations of public environments [15]. From this perspective, social acceptance concerns the perceived quality and social adequacy of the interaction experience itself, rather than an individual's intention to use a technology [16] or broader organizational or societal attitudes toward technological adoption [14].

The objective of the present study is to evaluate the user perception and social acceptance of an XR-Robot RA deployed in a real-world public environment. A user study has been conducted in a technological center, where participants interacted with an XR-based RA (see Fig. 1) and subsequently, but on demand, received navigation guidance from a mobile robot. Two experimental sessions were defined, each corresponding to a different navigation strategy, enabling a comparative analysis of how robot motion behavior influences perceived safety, comfort, and naturalness of interaction. Subjective feedback was collected through a structured questionnaire to assess user experience and social acceptance.

The remainder of this paper is structured as follows. Section II sheds light on the main contributions of this work, positioning them with respect to the related literature. Section III presents the proposed XR-Robot RA architecture, including the interaction process and the functional decomposition into interaction-centric and mobility-centric components. Section IV describes the user study conducted in a real-world public environment, together with the adopted evaluation methodology. Section V reports and discusses the results obtained from the user questionnaire. Concluding remarks and future developments are reported in Section VI.

II. Related Works and Contributions of the Paper

A. Related Works

RAs have been widely studied within the field of service robotics, with most contributions focusing on physically embodied robotic systems operating in public and human-shared environments [17]. Previous works have investigated robotic reception systems capable of greeting users, delivering information, supporting authentication procedures, and providing spatial guidance to target locations, while addressing challenges related to perception, dialogue management, and autonomous navigation [2], [4]. Several studies have emphasized socially aware navigation strategies and interaction mechanisms aimed at minimizing discomfort during motion in populated spaces [10], [18]. Additional research has explored user engagement techniques based on intention detection [7], multimodal communication [8], and behavioral adaptation to different user profiles (e.g., adults, children, elderly, or impaired individuals), to improve user engagement and acceptance while satisfying social, functional, and ethical constraints [9]. Despite significant progress in socially aware navigation and interaction design, the majority of robotic RAs rely on a single embodied robot responsible for both interaction and mobility, an architectural choice that can constrain interaction quality and system flexibility in crowded or operationally complex environments [3].

Like in the case of robotic solutions, virtual and XR-based agents have been investigated as reception and information interfaces in public and semi-public environments, i.e., spaces with controlled access and defined user groups, including healthcare facilities, educational institutions, workplaces, and assisted living settings [19]. Embodied Conversational Agents (ECAs) displayed on kiosks or large screens, together with XR avatars deployed through AR or MR devices, have demonstrated the ability to sustain natural language dialogue and expressive non-verbal communication via facial expressions, gaze, and gestures [20], [21]. Advances in Large Language Models (LLMs) and expressive rendering techniques have further enhanced conversational realism and personalization [22]. User studies have reported positive effects on perceived usability, trust, and engagement, including the ability of target users to independently use the system and receive valuable information [19], [23]. However, XR-based RAs are typically stationary and lack physical embodiment, restricting the ability to provide spatial guidance, accompany users, or operate beyond kiosk-based interaction, and limits operation in extended physical environments.

To overcome the limitations of purely robotic and purely virtual RAs, a smaller body of work has explored hybrid architectures that combine physical robots and virtual agents as complementary embodiments of a single intelligent system. Existing hybrid approaches have been investigated in

domains, including healthcare, education, and assistive interaction, where virtual agents provide expressive communication or cognitive support while robots deliver physical presence and task execution. Representative examples include cloud-based long-term interaction frameworks integrating a physical NAO humanoid robot (i.e., a commercially available platform developed by SoftBank Robotics) and a virtual avatar displayed on a tablet, sharing a common ontology and user model to support continuity and personalization across clinical and domestic settings [24], as well as systems combining mobile ECAs running on smartphones with humanoid robots acting as physical coaches for guided user interaction and digital training among older adults [25]. Other works have proposed tighter integration between robots and virtual assistants, enabling the coordinated execution of digital and physical tasks through shared representations and reasoning mechanisms. Representative approaches include architectures that couple LLM-powered virtual assistants with customized mobile robots equipped with sensing capabilities to support joint execution of cyber and physical tasks [22], as well as systems that embed three-dimensional ECAs into humanoid robots to exchange non-verbal cues in real-time and improve engagement, perceived social presence, and interaction [26].

Although the aforementioned studies highlight the potential benefits of combining virtual expressivity with robotic embodiment, reception services in open public environments remain under-explored, particularly with respect to systematic evaluation of user perception and social acceptance, and with respect to the impact of functional role separation between interaction-centric and mobility-centric components on perceived safety, comfort, and naturalness of interaction.

B. Contributions of the Paper

Motivated by the limitations identified in the literature on robotic, XR-based, and hybrid RAs, we propose an XR-Robot RA designed for operation in open public environments. The proposed approach follows a user-centered design rationale and originates from an analysis of the receptionist-user interaction process, including user approach and intention detection, information request, authentication, and task execution phases. Each phase introduces distinct technical, interactional, and social requirements related to perception, communication, responsiveness, safety, and compliance with social norms, characterizing public reception scenarios.

To address heterogeneous user needs and operational constraints, the proposed work introduces a process-driven XR-Robot RA architecture based on a functional decomposition of a receptionist's role. Interaction activities requiring expressive communication, dialogue management, and user identification are assigned to an interaction-centric XR agent, while activities involving physical movement and spatial guidance are handled by a mobility-centric robotic platform. Functional separation reflects user-centered considerations aimed at improving the clarity of interaction and preventing interference between communication processes and navigation behavior in human-populated environments.

The applicability of the proposed architecture is demonstrated by its deployment in a real-world technological center

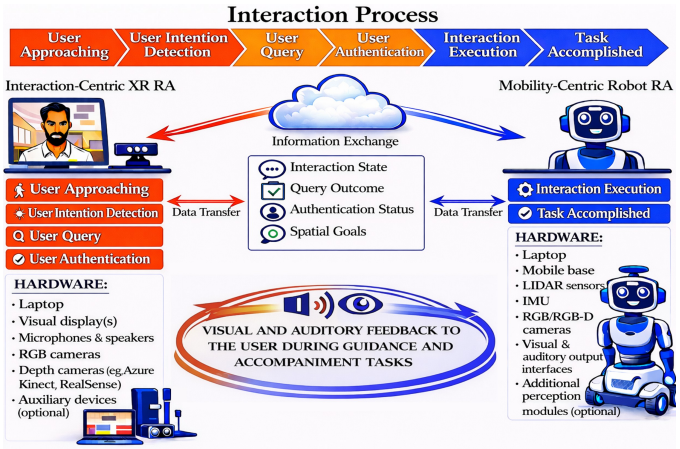

Fig. 2. Process-driven architecture of the proposed XR-Robot RA. The top bar illustrates the six phases of the receptionist-user interaction process. The interaction-centric XR RA (on the left) manages early interaction phases through multimodal perception and dialogue, while the mobility-centric robot RA (on the right) handles physical execution tasks. High-level interaction and task information are exchanged between the two components, enabling continuous audio-visual feedback to the user during guidance and accompaniment.

characterized by heterogeneous users and dynamic social activity. Operation in a public environment enables evaluation under realistic conditions and supports analysis of system behavior during real reception tasks.

A user study is conducted with the unified aim of assessing the XR-Robot RA's social acceptance in human-shared public spaces, using a structured questionnaire that evaluates perceived usefulness, ease of use, enjoyment, trust, safety, and overall user experience, while also comparing different robot navigation strategies. By using the same questionnaire to measure both the RA's qualities and users' responses to its motion behaviors, the study provides empirical insights into how navigation strategies influence perceived adaptiveness and social presence, and how hybrid embodiment together with user-centered functional role separation shape acceptance.

III. Proposed XR-Robot RA Architecture

The proposed XR-Robot RA architecture (see Fig. 2) is defined through a process-driven approach, in which the design of the agent follows from an explicit modelling of the receptionist-user interaction. Reception services in public and semi-public environments are commonly described as structured interactions comprising a defined set of phases, independently of the specific deployment context or application domain [2], [4], [5], [17]. The formalization of the interaction process enables the systematic identification of the functional capabilities required by the RA and provides a principled basis for its architectural design.

The interaction process considered in this work models the progression of a receptionist-user interaction from the initial encounter to the completion of the requested service, as summarized in Fig. 2. The process starts with a *User Approaching* phase, in which a user enters the operational area and approaches the reception point, requiring continuous perception of human presence within the environment by the receptionist. Public and semi-public spaces exhibit continuous human activity that does not necessarily correspond to an

intention to interact with the receptionist; therefore, an explicit *User Intention Detection* phase is required to differentiate between generic human presence and an explicit intention to initiate interaction. Intention detection relies on multimodal social cues, including spatial proximity, body orientation, gaze direction, gestures, and verbal signals, as commonly adopted in real-world deployments of service robots and interactive agents [6], [7]. Following detection of interaction intent, the process advances to the *User Query* phase, during which the user initiates a dialogue and communicates a request to the RA. This phase represents an explicit entry point of the interaction and enables task specification. Depending on contextual and security requirements, the interaction process may include a *User Authentication* phase. Authentication is treated as an optional interaction phase whose activation depends on domain-specific access policies rather than architectural assumptions and may involve different verification mechanisms, ranging from lightweight authorization checks to structured identity verification procedures [4], [27]. After authentication, when required, the interaction proceeds to the *Interaction Execution* phase. In this phase, RA delivers the requested service, either through information provisioning or through physical actions within the environment, such as escorting the user to a target location. The completion of service delivery marks the *Task Accomplished* phase.

Analysis of the interaction process reveals heterogeneous functional requirements across different phases. The phases preceding the execution of the service emphasize the perception of social cues, multimodal communication, and dialogue management, aspects central to the design of ECAs and multimodal interactive systems [12], [13], [19], [20]. In contrast, phases that involve physical guidance impose requirements related to autonomous navigation, environment perception, and socially compliant motion in human-populated spaces, as discussed in the literature on service robotics [3], [10], [18]. Satisfying these requirement categories within a single embodiment often leads to compromises between interaction quality and navigation performance.

Based on the functional analysis of the receptionist-user interaction process, the proposed architecture decomposes RA into two cooperating components: an interaction-centric agent and a mobility-centric agent. The decomposition reflects the distinct requirements associated with interaction-oriented phases and phases involving physical action within the environment, while preserving a coherent receptionist-user workflow. The resulting architecture supports modular configuration across different application scenarios, allowing interaction phases and hardware components to be selectively enabled or omitted according to the contextual requirements, while maintaining a clear functional separation between interaction-centric and mobility-centric capabilities. The interaction flow can be supported by different hardware configurations, as illustrated in Fig. 2, which define a general architecture suitable for implementing receptionist scenarios under realistic conditions.

A. Interaction-centric Receptionist Agent

The interaction-centric RA is designed as an embodied XR-based receptionist deployed at a fixed reception point. The agent is presented to users through a visual interface, typically a large display, on which a three-dimensional virtual body or avatar is rendered to provide social presence and support expressive interaction. Using virtual embodiment allows the agent to convey social cues, such as gaze direction, facial expressions, and gestures, contributing to a more natural and engaging interaction experience. The XR-based RA manages interaction phases that do not require physical displacement within the environment, including user presence monitoring, intention detection, query handling, and authentication. Deployment at a fixed location allows controlled spatial configuration of sensors and interaction interfaces, supporting a consistent user perception and engagement during dialogue-based interaction in public or semi-public environments.

Hardware supporting the interaction-centric RA is selected to enable robust multimodal perception and communication. Mandatory components include visual output devices for information presentation and feedback, microphones and speakers for speech-based interaction, and visual sensing devices for perception of users and the surrounding environment. RGB cameras support user detection and visual analysis, while depth-capable sensors provide complementary spatial information that improves the robustness of user localization and intention detection in indoor settings. Depth sensing is commonly realized through RGB-D cameras or dedicated depth sensors based on time-of-flight or structured-light technologies, which are widely adopted in HRI and embodied agent research to support robust user detection, tracking, and interpretation of non-verbal cues in indoor environments [7], [12], [19]. Representative devices include commercially available platforms, such as the Azure Kinect DK [28] and the Intel RealSense D435 [29]. Depth information supports estimation of user distance and relative position during the *User Approaching* phase and facilitates distinction between interacting users and background elements during intention detection. Compared to RGB-only perception, depth sensing improves robustness under variable illumination conditions and supports tracking of multiple users in crowded environments. Computational resources associated with the XR-based RA support dialogue management, perception processing, and coordination of multimodal interaction. Additional auxiliary devices, such as external displays or supplementary cameras, may be integrated to enhance robustness or scalability while preserving the functional role of the agent.

B. Mobility-centric Receptionist Agent

The mobility-centric RA is realized as a physically embodied mobile service robot operating in human-shared indoor environments. The agent is conceived as a mobile platform, potentially humanoid or human-inspired, equipped with a mobile base and interaction interfaces that allow continuous engagement with users during movement. Physical embodiment and mobility enable spatial interaction beyond the reception point and support navigation-based services, such as escorting users to offices, exhibition areas, or service locations. The robotic RA manages interaction phases that require physical action within the environment, with a primary focus on user guidance and accompaniment tasks.

Hardware supporting the mobility-centric RA is selected to ensure reliable autonomous operation and safe interaction in human-shared indoor environments. Core components include a locomotion system suitable for indoor navigation, typically realized through a mobile base enabling smooth and controllable motion in constrained spaces, together with sensing capabilities required for perception of both static environmental elements and dynamic agents. The locomotion system supports socially compliant motion, allowing adaptation of speed, trajectory, and stopping behavior during user guidance and accompaniment tasks. Sensing capabilities for autonomous navigation involve localization and mapping sensors, such as two-dimensional or three-dimensional laser scanners and inertial measurement units, which support accurate state estimation, obstacle detection, and environment representation, and are widely adopted in socially aware navigation systems [3], [10], [30]. Laser-based perception provides reliable range measurements for navigation and obstacle avoidance, while inertial sensing contributes to motion estimation and robustness under temporary perception degradation. Visual perception devices, including RGB or RGB-D cameras mounted on the robot body or head, complement range-based sensing by enabling detection and tracking of surrounding agents and obstacles and by supporting monitoring of a user's position relative to the robot during guidance tasks. Visual perception enables adaptive behaviors, including speed modulation, waiting strategies, and re-engagement when deviations from the expected user-following position are detected. Visual and auditory output interfaces, including screens, speakers, light signals, or expressive head movements, enable communication with the user during navigation, supporting delivery of instructions, confirmations, and feedback throughout the guidance process, as recommended in prior work on robotic guidance and wayfinding systems [2], [5]. Computational resources onboard the mobile platform support real-time processing for navigation, perception, and coordination of interaction-related outputs. Additional perception modules, including rear-mounted cameras or range sensors, may be integrated to support user-following behavior and to improve robustness in crowded or highly dynamic environments, enabling verification of user presence and relative position during accompaniment tasks.

C. Coordination of Interaction-centric and Mobility-centric Receptionist Agents

Coordination between the interaction-centric RA and the mobility-centric RA is organized as a structured handover within the receptionist-user workflow. The former, implemented as an XR-based agent, serves as the primary contact point during the initial interaction phases, managing user engagement, intention interpretation, query handling, and task definition. When execution of the requested service requires physical action within the environment, responsibility for service delivery transfers to the latter, which assumes control of navigation-based activities while preserving continuity of user engagement until task completion.

Information flow between the two components of the RA relies on the transmission of high-level interaction state and task parameters, including query outcomes, authentication status (when applicable), and spatial goals associated with the requested service. Coordinated operation requires hardware support for reliable and low-latency data transmission between the computational units associated with the stationary and mobile agents. Network interfaces integrated within both agents enable continuous data transfer during task execution, while adequate computational resources support maintenance of a consistent interaction across components. Visual and auditory interfaces on both agents contribute to interaction continuity by providing feedback aligned with the current execution phase.

During guidance and accompaniment tasks, the mobility-centric agent maintains user engagement via visual and auditory feedback while monitoring user position and progression toward the target destination. Completion of the navigation-based task concludes the interaction session and restores the system to an idle reception state. Separation of interaction-centric and mobility-centric responsibilities, supported by computational and data transmission resources, defines a coherent architectural organization for receptionist services.

IV. Description of Tests

A. User Study

The XR-Robot RA developed in this work was validated through a user study conducted in a real-world technological center open to the general public. Experiments were conducted during regular opening hours in a semi-structured indoor space characterized by corridors, four predefined destination points corresponding to service and meeting locations within the center, and continuous human activity. The environment included static obstacles, such as tables, chairs, and exhibition structures, as well as dynamic obstacles mainly represented by people moving within the area. The scenario reflects typical operating conditions for an RA deployed in public spaces.

The setup supports all phases of the receptionist-user interaction process defined in Section III, namely *User Approaching*, *User Intention Detection*, *User Query*, *User Authentication*, *Interaction Execution*, and *Task Accomplished*, through an XR-Robot hardware architecture comprising an interaction-centric agent and a mobility-centric robotic platform.

The *User Approaching* and *User Intention Detection* phases were supported by the interaction-centric XR RA deployed at a fixed reception point. Hardware components included a laptop (Dell XPS 17 9730) equipped with a 13th Gen Intel Core i9-13900H CPU, 64 GB of RAM, and an NVIDIA GeForce RTX 4080 GPU, together with a Microsoft Azure Kinect DK camera and a standard RGB camera. Depth sensing provided by the Azure Kinect enabled monitoring of human presence, estimation of user distance, and spatial configuration in the reception area, supporting the distinction between generic human activity and explicit intention to interact. The RGB camera was used for visual acquisition during subsequent interaction phases, including user authentication. Audio input was provided using a RØDE VideoMic GO directional microphone [31] to capture verbal cues, while a visual display connected to the laptop enabled multimodal interaction by presenting graphical and textual information to the user.

The *User Query* and *User Authentication* phases were implemented using the Socially-acceptable Extended Reality Models And Systems (SERMAS) Toolkit [32], integrated with the same interaction-centric hardware configuration. Voice-based dialogue was supported through the aforementioned directional microphone, while audio output was provided through the laptop-integrated speakers based on Intel® Smart Sound Technology for MIPI SoundWire®. QR-code-based authentication was enabled via the RGB camera. The adopted configuration allowed multimodal interaction combining speech, visual feedback, and identity verification, providing the information required to define the requested service and associated execution parameters.

The *Interaction Execution* and *Task Accomplished* phases were performed by a mobility-centric robotic platform, implemented using the TIAGo Pro mobile manipulator developed by PAL Robotics [33]. The robot was connected to an external laptop equipped with an Intel Ultra 7 CPU, 16 GB of RAM, and an NVIDIA RTX 4050 GPU, running Ubuntu 22.04 and ROS 2 Humble. The robotic platform features an omnidirectional base with four wheels, an onboard computer (Intel i5 CPU, 8 GB RAM, 250 GB SSD, Wi-Fi 6, and Bluetooth 5.2), two 360° LIDAR sensors, a 6-DoF IMU, and stereo speakers for audio feedback. A head-mounted display was used to provide visual information and feedback to users during guidance and accompaniment.

Communication between the interaction-centric XR RA and the mobility-centric robot RA was established through a dedicated Ethernet SSH connection, ensuring stable and low-latency exchange of interaction state information, including query outcomes, authentication status, and spatial goals. To support perception during navigation and accompaniment, a ZED2 AI stereo camera [34] was mounted on the rear expansion panel of the robot via a custom 3D-printed support and connected to the external laptop through a USB 3.1 interface. The camera was employed to measure the distance between the human and the robot, and to support user detection and tracking during motion.

B. Test Protocol

The experimental protocol evaluated the receptionist-user interaction process described in Section III through a single end-to-end interaction supported by the XR-Robot RA. All participants completed one individual session, defined as a continuous interaction from the reception point to arrival at the assigned destination.

Two experimental sessions were conducted, each with a distinct navigation mode of the mobility-centric robot:
– *Mode 1 (NAV2-based navigation):* The first experimental session employed the NAV2 framework [35], the standard ROS 2 navigation stack, providing modular components for localization, global and local path planning, and obstacle avoidance in indoor environments;
– *Mode 2 (CBF-based STL navigation):* The second experimental session employed a navigation framework based on Control Barrier Functions (CBF) and Signal Temporal Logic (STL) specifications, following the approach presented in [36]. The framework enforces formally defined safety and social constraints during motion planning, enabling navigation behaviors that are compliant with temporal and spatial specifications in shared environments.

The use of two navigation modes enabled a comparative evaluation of user perception under different robot motion behaviors, while preserving identical interaction procedures and environmental conditions.

Before the session, participants received a personalized QR code via a web link. The QR code acted as a digital access token associated with a scheduled visit within the technological center and encoded one of the four predefined destination points. The code was used during the *User Authentication* phase of the interaction [27]. At the beginning of the session, participants received information about the study and procedure, then approached the reception point, initiating the *User Approaching* phase. The interaction-centric XR RA detected human presence and identified the intention to interact through the *User Intention Detection* phase, based on proximity and behavioral cues. The system entered the *User Query* phase and engaged participants through spoken dialogue to collect the user request. The interaction then progressed to the *User Authentication* phase, in which participants presented the QR code. The code was acquired through the RGB camera and verified by the system, enabling retrieval of the destination associated with each participant. After authentication, the assigned destination was communicated through multimodal feedback combining dialogue and visual information. Participants were then invited to proceed with the guidance service. Upon acceptance, the *Interaction Execution* phase began, and the mobility-centric robot guided participants from the reception area to the assigned destination, according to the navigation mode associated with the experimental session. During navigation, the robot monitored the relative distance between the robot and the participant, and provided verbal feedback when deviations from the expected following distance were detected, inviting the participant to move closer or farther as appropriate. The interaction concluded with the *Task Accomplished* phase upon arrival at the destination.

For safety purposes, a trained staff member continuously supervised the robot during navigation while holding the robot's emergency stop device, allowing immediate interruption of motion if required.

To further support the qualitative assessment of the proposed XR-Robot RA, a demonstration video recorded during the user study is provided as supplementary material (see extension 1). The video shows the complete receptionist-user interaction, from XR-based dialogue to robot guidance (mode 2), including adaptive feedback, safe navigation, and guidance to predefined destinations under kinematic constraints.

C. Test Participants

Participants were recruited on a voluntary basis and received detailed information about the study objectives, procedure, and interaction with the XR-Robot RA. Written informed consent was obtained from all of them, and the right to withdraw at any time was guaranteed. Data were collected in anonymized and

TABLE I
DEMOGRAPHIC INFORMATION OF PARTICIPANTS.

Category	Sub-category	Group 1 (N=20)	Group 2 (N=25)
Gender	Men	65%	24%
	Women	35%	76%
	Non-binary	-	-
	Transgender	-	-
	Intersexual	-	-
	Prefer not to say	–	-
Age Range	18-35 years	100%	72%
	36-50 years	0%	4%
	51-65 years	0%	24%
	65+ years	-	-

aggregated form, limited to interaction-related feedback, with no personally identifiable or sensitive information retained. The study complied with institutional guidelines on ethics, privacy, and data protection.

A total of 45 participants were involved in the experimental validation and assigned to two independent groups, each associated with a different navigation mode of the mobility-centric robot. For simplicity, participants testing the XR-Robot RA with NAV2-based navigation are hereafter referred to as group 1, while those testing the CBF-based STL navigation mode are referred to as group 2. The two groups comprised 20 and 25 participants, respectively, each interacting with the system under their assigned navigation framework. Each participant was assigned to a single experimental condition and exposed to only one robot navigation behavior.

All participants in both groups followed the same interaction procedure and operated in the same physical environment, with the mobility-centric robot navigation mode during the guidance phase as the only experimental variable. This between-subjects design enabled comparison of user perception across navigation behaviors while avoiding carry-over effects.

The demographic characteristics of the two groups of participants, including gender and age distribution, are reported in Table I.

D. Subjective Metrics

Questionnaire Development: The evaluation of the proposed XR-Robot RA was based on subjective metrics aimed at assessing users' perceptions of the receptionist–user interaction and the social acceptance of the robot during reception and guidance tasks. User perceptions were gathered through a structured questionnaire, developed by adapting an extended version [15] of the original Almere model [16]. The model's dimensions correspond to the features and capabilities that XR and robotic technologies must provide to support socially acceptable interaction experiences, as discussed in Section I. It was administered under identical experimental conditions to all participants, allowing for comparison of the two navigation modes of the mobility-centric robot.

The questionnaire included 16 attributes, each capturing a specific psychological factor of user perception related to social acceptance. The attributes are defined as follows:

- *Anxiety (ANX):* Degree of fear, discomfort, or intimidation experienced during interaction;
- *Attitude towards Technology (ATT):* Overall positive or negative evaluation of using the system;
- *Facilitating Conditions (FC):* Perceived availability of resources, knowledge, and support required to use the system;
- *Informational Privacy (IP):* Concerns related to the collection, misuse, or exposure of personal data;
- *Intention to Use (ITU):* Willingness to use the system in future interactions;
- *Perceived Adaptiveness (PA):* Perceived ability of the system to adapt to user needs and context;
- *Perceived Enjoyment (PE):* Level of engagement, enjoyment, and interest during interaction;
- *Perceived Ease of Use (PEOU):* Perceived effort required to use the system;
- *Physical Privacy (PP):* Perception of respect for physical and spatial boundaries;
- *Perceived Sociability (PS):* Perceived capability to exhibit appropriate social behaviors;
- *Perceived Usefulness (PU):* Perceived usefulness in supporting reception and guidance tasks;
- *Safety (SAF):* Perception of physical safety and absence of risk during interaction;
- *Security (SEC):* Confidence in authentication mechanisms and protection against unauthorized access;
- *Social Influence (SI):* Perceived influence of others on the decision to use the system;
- *Social Presence (SP):* Perception of interacting with a socially present entity;
- *Trust (TRU):* Confidence in reliability and in the system acting in the user's interest.

The attribute set builds upon [15] by introducing PP and SAF to account for the presence of a mobility-centric robotic component operating in shared physical spaces. Spatial intrusion and safety risks make PP and SAF relevant dimensions of user experience, as also highlighted in previous works (see, e.g., [37]). In addition, SEC-related questions were extended to assess aspects of QR code-based user authentication.

Each attribute was measured through multiple questionnaire items obtained by grouping related questions (see Table II), for a total of 57 entries, each designed to reflect specific aspects of the receptionist–user interaction. Participants rated each item on a 5-point Likert scale ranging from 1 to 5 (1 = "totally disagree", 2 = "disagree", 3 = "don't know", 4 = "agree", 5 = "totally agree"). Items were presented in random order and alternated between positively and negatively worded statements to improve response reliability [38].

Data Analysis: Data were processed in anonymized and aggregated form to assess differences in perceived interaction quality and social acceptance between the two navigation modes. For each attribute, a mean score was computed by averaging the responses to its associated items, with a value of 3.00 considered neutral and values above and below indicating positive and negative perceptions, respectively.

All statistical analyses were conducted with IBM SPSS Statistics (version 30) [39]. Data normality was assessed

TABLE II
QUESTIONNAIRE ITEMS GROUPED BY ATTRIBUTES FOR THE ASSESSMENT OF USER PERCEPTION AND SOCIAL ACCEPTANCE.

Attribute	Question
ANX	1. If I were to use the XR-Robot RA, I would be afraid to make mistakes with it. 2. I find the XR-Robot RA scary. 3. I find the XR-Robot RA intimidating.
ATT	4. I think it is a good idea to use the XR-Robot RA as a receptionist. 5. I think the XR-Robot RA makes check-in procedures (e.g., in hotels, offices, etc.) more interesting. 6. It is good to make use of the XR-Robot RA.
FC	7. I have everything I need to use the XR-Robot RA. 8. I know enough about the XR-Robot RA to make good use of it.
IP	9. I think the XR-Robot RA could over-collect information. 10. It is highly probable that I will share my personal information with the XR-Robot RA. 11. I am concerned about the potential exposure of sensitive data through the XR-Robot RA to unauthorized parties. 12. I am concerned that the XR-Robot RA may leak my personal information to irrelevant third parties. 13. I am concerned that using the XR-Robot RA could result in the misuse of my personal information.
ITU	14. I think I will use the XR-Robot RA during the check-in activity. 15. I am certain to use the XR-Robot RA during the check-in activity. 16. I plan to use the XR-Robot RA during the check-in activity.
PA	17. I think the XR-Robot RA can be adaptive to what I need. 18. I think the XR-Robot RA will only do what I need at that particular moment. 19. I think the XR-Robot RA will help me when I consider it to be necessary.
PE	20. I enjoy the XR-Robot RA talking to me. 21. I enjoy doing things with the XR-Robot RA. 22. I find the XR-Robot RA enjoyable. 23. I find the XR-Robot RA fascinating. 24. I find the XR-Robot RA boring.
PEOU	25. I think I will know quickly how to use the XR-Robot RA. 26. I find the XR-Robot RA easy to use. 27. I think I can use the XR-Robot RA without any help. 28. I think I can use the XR-Robot RA when there is someone around to help me. 29. I think I can use the XR-Robot RA when I have a good manual.
PP	30. The mobility-centric RA is entering areas in the building that it should not access.
PS	31. I consider the XR-Robot RA pleasant to have a conversation with. 32. I find the XR-Robot RA pleasant to interact with. 33. I feel the XR-Robot RA understands me. 34. I think the XR-Robot RA is nice.
PU	35. I think the XR-Robot RA is useful to me during the check-in procedures. 36. It would be convenient for me to have the XR-Robot RA during the check-in activities. 37. I think the XR-Robot RA can help me with different activities.
SAF	38. I do not feel safe due to the presence of the mobility-centric RA. 39. Overall, I perceive the mobility-centric RA as risky.
SEC	40. I am concerned about the potential for unauthorized users to access the XR-Robot RA. 41. I understand the purpose of the QR code-based authentication process. 42. I am aware of how the QR code is used to verify my identity. 43. I am satisfied with the authentication procedure.
SI	44. I think the organization's staff would like me to use the XR-Robot RA. 45. I think it would give a good impression if I use the XR-Robot RA.
SP	46. When interacting with the XR-Robot RA, I felt like I am talking to a real person. 47. It sometimes felt as if the XR-Robot RA was really looking at me. 48. I can imagine the virtual avatar to be a living creature. 49. I often think the XR-Robot RA is not a real person. 50. Sometimes the XR-Robot RA seems to have real feelings.
TRU	51. I would trust the XR-Robot RA if it gave me advice. 52. I would follow the advice and trust the information the XR-Robot RA gave me. 53. I believe that only legitimate individuals can access the XR-Robot RA. 54. I am sure that the XR-Robot RA is maintaining a secure interaction environment. 55. I am confident that my anonymity is protected by the XR-Robot RA. 56. I believe that the XR-Robot RA acted in my best interest. 57. The XR-Robot RA performed its role of offering services really well.

using the Shapiro-Wilk test [40]. Results indicated that several factors were not normally distributed and that homogeneity of variance was not consistently satisfied. Therefore, the differences between the two independent groups were evaluated using the Mann-Whitney U test (also known as the Wilcoxon rank-sum test) [41], which does not assume normality and is appropriate for the collected measurements. All analyses were conducted using a two-tailed approach, with statistical significance set at $p < .05$ [42].

V. RESULTS

The results are reported to evaluate the effectiveness of the proposed XR-Robot RA architecture in terms of user perception and social acceptance during the receptionist-user interaction. To provide an overall assessment of the system independently of the navigation mode, Table III reports the aggregated mean scores and Standard Deviations (SD) across all participants. Table IV then presents the results separately for the two groups, while Table V reports the outcomes of the Mann-Whitney U test used to assess statistically significant differences between them.

A. Overall User Perception

The aggregated results across all participants (see Table III) show that mean scores exceed the neutral threshold (i.e., 3.00) for most attributes, indicating a generally positive level of user acceptance of the XR-Robot RA architecture. Attributes (e.g., ATT and PE) associated with the interaction-centric components received high scores, suggesting that participants positively perceived the XR interface and the virtual avatar. Attributes related to usability and interface design (i.e., PEOU, IP, and SEC) were also rated favorably, with the exception of PEOU, which exhibits relatively lower scores, highlighting some limitations related to usability. Reported difficulties are mainly associated with the clarity of the voice-based interaction and the readability of the graphical interface, indicating the need for improvements, particularly in multimodal communication and interface accessibility. Mobility-centric attributes (i.e., PP and SAF), together with low ANX scores and high TRU scores, indicate that the system is perceived as safe and reliable, with a high level of participants' trust, supporting its suitability for deployment in public or semi-public environments where user comfort is essential. Similarly, attributes such as PU and ITU received high scores, indicating that the XR-Robot RA is perceived as useful for reception activities, reinforcing its applicability. In particular, in the Almere model, ITU is considered a proxy of social acceptance [43]. In contrast, SP values remain lower than other attributes, indicating that the XR-Robot RA is not perceived as strongly human-like, which is consistent with the hybrid nature of the system, where interaction is distributed between a virtual avatar and a physical robot rather than embodied in a single anthropomorphic entity.

B. Comparison between Navigation Modes

To further investigate the impact of robot motion behavior, results are analyzed separately for the two navigation modes

TABLE III
MEAN SCORES AND STANDARD DEVIATIONS FOR ALL ATTRIBUTES (AGGREGATED RESULTS).

Attribute	Mean	SD	Attribute	Mean	SD
ANX*	4.19	0.71	PP	4.67	0.95
ATT	3.79	0.84	PS	3.88	0.79
FC	4.09	0.73	PU	4.21	0.85
IP	3.50	0.96	SAF	4.56	0.91
ITU	4.02	1.03	SEC	4.33	0.57
PA	3.03	0.48	SI	3.89	0.95
PE	4.04	0.69	SP	2.35	0.76
PEOU	3.60	0.52	TRU	4.01	0.52

* A high mean score for ANX indicates a positive outcome, i.e., low measured ANX

TABLE IV
MEAN SCORES AND STANDARD DEVIATIONS FOR ALL ATTRIBUTES, REPORTED SEPARATELY FOR THE TWO GROUPS. ARROWS INDICATE CHANGE FROM GROUP 1 TO GROUP 2: ↑ INCREASE, ↓ DECREASE, → NO CHANGE. PERCENTAGES SHOW RELATIVE CHANGE.

Attribute	Mean		SD	
	Group 1	Group 2	Group 1	Group 2
ANX*	4.23	4.16 ↓ (-1.65%)	0.87	0.57 ↓ (-34.48%)
ATT	3.57	3.97 ↑ (11.20%)	0.85	0.80 ↓ (-5.88%)
FC	3.95	4.20 ↑ (6.33%)	0.72	0.72 → (0.00%)
IP	3.23	3.71 ↑ (14.86%)	0.84	1.00 ↑ (19.05%)
ITU	3.75	4.24 ↑ (13.07%)	1.13	0.90 ↓ (-20.35%)
PA	2.85	3.17 ↑ (11.23%)	0.55	0.36 ↓ (-34.55%)
PE	3.85	4.19 ↑ (8.83%)	0.58	0.75 ↑ (29.31%)
PEOU	3.63	3.57 ↓ (-1.65%)	0.41	0.60 ↑ (46.34%)
PP	4.60	4.72 ↑ (2.61%)	1.10	0.84 ↓ (-23.64%)
PS	3.69	4.03 ↑ (9.21%)	0.86	0.72 ↓ (-16.28%)
PU	4.03	4.35 ↑ (7.94%)	1.02	0.67 ↓ (-34.31%)
SAF	4.60	4.52 ↓ (-1.74%)	0.82	0.98 ↑ (19.51%)
SEC	4.34	4.33 ↓ (-0.23%)	0.60	0.56 ↓ (-6.67%)
SI	3.60	4.12 ↑ (14.44%)	0.99	0.86 ↓ (-13.13%)
SP	2.05	2.59 ↑ (26.34%)	0.76	0.68 ↓ (-10.53%)
TRU	3.89	4.11 ↑ (5.66%)	0.60	0.43 ↓ (-28.33%)

* a high mean score for ANX indicates a positive outcome, i.e., low measured ANX

(see Table IV). Group 2 consistently exhibits higher mean scores across nearly all attributes, indicating more favorable participants' perceptions associated with the corresponding navigation mode. Statistically significant differences are observed for PA (U = 161.50, $p = 0.04$) and SP (U = 147.50, $p = 0.02$), with higher values in group 2 (see Table V), indicating enhanced perceived adaptiveness and social engagement. No significant differences are observed for attributes such as ATT, FC, ITU, PE, PS, PU, SI, and TRU, indicating similar perceptions of usefulness, trust, and task suitability across both navigation strategies.

C. Variability of Responses

Looking at the aggregated results (see Table III), SD remains relatively low across most attributes, indicating a consistent overall perception of the XR-Robot RA architecture among participants. When analyzing the two navigation modes separately (see Table IV), variability remains limited across both groups, confirming consistency of participants' evaluations. Notably, for PA, SD decreases from 0.55 in group 1 to 0.36 in group 2, indicating a more homogeneous perception of adaptiveness under the CBF-based STL navigation mode.

TABLE V
INDEPENDENT SAMPLES NON-PARAMETRIC MANN–WHITNEY U TEST RESULTS FOR ALL ATTRIBUTES.

Attribute	U	*p*	Attribute	U	*p*
ANX	211.50	.37	PP	250.50	1.00
ATT	186.00	.14	PS	196.00	.21
FC	194.50	.19	PU	215.00	.41
IP	177.00	.10	SAF	264.50	.70
ITU	179.50	.10	SEC	256.50	.89
PA	161.50	.04*	SI	175.50	.08
PE	167.50	.06	SP	147.50	.02*
PEOU	247.00	.95	TRU	197.50	.23

* denotes statistically significant differences ($p < .05$)

D. Summary of Findings

Considering all participants, the proposed XR-Robot RA architecture is positively evaluated in terms of interaction quality and social acceptance, regardless of the navigation mode. Results can be interpreted in light of the Almere model, where social acceptance is primarily reflected by ITU and is influenced by functional, social, and affective factors. High scores for PU, TRU, and ATT indicate that the system is well accepted and is perceived as useful, reliable, and suitable for reception tasks. Low ANX, together with high SAF and SEC, further reinforces users' trust and reduces potential barriers to acceptance in public deployment contexts. Lower PEOU values highlight usability limitations that may affect full adoption, suggesting the need for improvements in interaction design. Within this overall positive assessment, differences between navigation modes highlight the contribution of robot motion behavior to interaction quality. In particular, higher PA and stronger perceived SP support user engagement and positively influence ITU. Therefore, results indicate that, beyond functional performance, adaptive and socially aware behaviors, particularly those influenced by navigation strategies, play a key role in shaping social acceptance and enhancing interaction quality in XR-Robot RAs.

VI. CONCLUSIONS

This paper presented an Extended Reality-Robot Receptionist Agent designed through a functional decomposition of interaction-centric and mobility-centric roles. The proposed architecture enables the separation of communication and navigation tasks, supporting coherent and socially acceptable interaction in public environments. The results from the user study conducted in a real-world setting indicate an overall positive user perception and evaluation of the system's social acceptance. The observed outcomes can be attributed to the hybrid architecture underlying the system, which effectively supports reception tasks, as well as to the robot's navigation behavior, which influences social acceptance attributes.

Future work will focus on improving interaction clarity and usability, particularly through multimodal input guidance. Interface refinements, including adaptive visual layouts, will aim to support accessibility across diverse users and environments. Additional developments will aim to strengthen system's expressiveness by integrating richer social cues and multimodal

feedback, while further evaluations in more complex settings will assess scalability and robustness.

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